



CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

INFORMATION TECHNOLOGY

Paper 03/2 – General Proficiency

ALTERNATIVE TO SBA

1 hour 30 minutes

17 JANUARY 2011 (a.m.)

INSTRUCTIONS TO CANDIDATES

- 1. This paper consists of FOUR questions.**
- 2. Answer ALL questions.**
- 3. Number EACH answer correctly in the answer booklet.**

1015

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1. **WORD PROCESSING**

Consider the following memo:

CENTRE FOR INFORMATION TECHNOLOGY

REPAIRS AND MAINTENANCE DEPARTMENT

FROM: Chief Technical Officer

TO: Paul Jones, Technical Support Officer of School of Medicine

DATE: November 25, 2010

SUBJECT: BACKING UP OF CRITICAL DATA

We have observed that the hard drives on a number of computers sent to be repaired have crashed due to problems related to power. As Technical Support Officer assigned to the various schools, it is critical that you put measures in place to backup important data. You should backup critical data on portable storage media such as CD-ROM, DVD-ROM, external hard drive or USB flash drive. The backup copies of data should be scanned for viruses and kept in a secure location offsite

Michael Smith

1 Graduate of MIT

Implementing New Technology

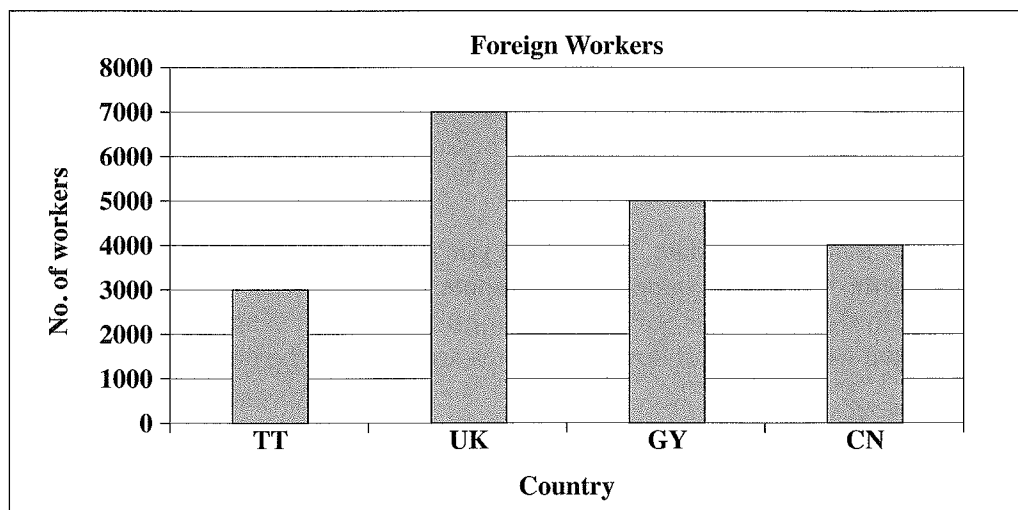
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- (a) What justification has been applied to the text in the body of the memo? (1 mark)
- (b) Identify TWO different line spacings used in the memo. (2 marks)
- (c) State THREE formatting features used in the memo. (3 marks)
- (d) The word 'bakup' is spelt incorrectly. What feature of the word processing program will you use to change all occurrences of 'bakup' to 'backup'? How many changes will be made? (2 marks)
- (e) Rather than type the date, how would you modify the memo to ensure that the date of the memo will be the same as the computer date? (3 marks)
- (f) Indicate which of the following features are found in the memo:
- Header
 - Footnote
 - Footer
 - Endnote
- (1 mark)
- (g) Explain how you would remove the underline in the Subject of the memo. (2 marks)
- (h) You wish to merge this memo to the FIVE other Technical Support Officers. Name the TWO documents which are required to perform a merge. (2 marks)
- (i) Write the text in the memo that should be replaced by merge fields. (2 marks)
- (j) State, with reasons, the number of merge fields this memo should have. (2 marks)

Total 20 marks

2. SPREADSHEET

Consider the following chart which was created from the data in a spreadsheet:



- Name the type of chart shown above. (1 mark)
- Write the name of another type of chart which would be suitable to represent the data. (1 mark)
- What is the purpose of the chart? (1 mark)
- State the chart and X-axis titles. (2 marks)
- The following table represents part of the spreadsheet. Draw this table in your answer booklet. Write the data in this table that were used to create the chart.

	A	B
1	Country	Worker
2		
3		
4		
5		

- (4 marks)
- What numeric data format would be appropriate for the data in the table you completed in Part (e)? (1 mark)
- Indicate the range of cells that you would use to sort ALL the data by the number of workers. (1 mark)
- Write the country that would be at the top of the list if the data were sorted in DESCENDING ORDER by the number of workers. (1 mark)

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- (i) Write a function based on the data in Part (e) to calculate the TOTAL number of foreign workers. (2 marks)
- (j) You wish to filter the data for all countries with over 4000 workers. Draw the table listed below in your answer booklet and write the criterion that would filter the data.

Country	Worker

(2 marks)

- (k) Write the results produced from the criterion given in Part (j). (1 mark)
- (l) You wish to insert a title to your spreadsheet such that
- it contains the text 'FOREIGN WORKERS – 2010'
 - it is in Row 1 and centered across the columns used by the spreadsheet.

Explain how you would do this using the spreadsheet created in Part (e). (3 marks)

Total 20 marks

3. DATABASE MANAGEMENT

Consider the following database with two tables.

TABLE 1: REGION

R_ID	R_NAME
CB	Caribbean
LA	Latin America
SA	South America
NA	North America
EU	Europe
AS	Asia
ME	Middle East
AF	Africa

TABLE 2: VISIT

R_ID	Type	Visit
NA	Diplomat	250
NA	Vacation	700
NA	Business	1500
AF	Diplomat	25
AF	Business	50
ME	Business	20
ME	Diplomat	10
AS	Diplomat	50
AS	Business	500
EU	Vacation	700
EU	Business	400
EU	Diplomat	200
SA	Diplomat	100
SA	Business	300
LA	Diplomat	25
LA	Business	50
CB	Vacation	2000
CB	Diplomat	500
CB	Business	1200

- (a) In your answer booklet, draw and complete the following table which represents the structure of the database table VISIT.

Field Name	Data Type

(3 marks)

- (b) State the number of records and fields in the REGION table. (2 marks)
- (c) Why is the **R_ID** field NOT suitable as a PRIMARY key in the VISIT table? (1 mark)
- (d) You are required to arrange the records in the VISIT table in **ascending** order by the **R_ID** field and then by the **Visit** field. Explain how you would do this. (2 marks)
- (e) The two tables have been joined on a common field; state the number of diplomats from Africa who visited the country. (1 mark)

- (f) The following report was created from the data in the database:

<i>VISITORS 2010</i>		
<i>R_ID</i>	<i>Visit</i>	<i>Type</i>
<i>AF</i>	25	Diplomat
	50	Business
<i>Summary for 'R_ID' = AF (2 detail records)</i>		
Sum	75	
<i>AS</i>	50	Diplomat
	500	Business
<i>Summary for 'R_ID' = AS (2 detail records)</i>		
Sum	550	
<i>CB</i>	500	Diplomat
	1200	Business
	2000	Vacation
<i>Summary for 'R_ID' = CB (3 detail records)</i>		
Sum	3700	
<i>EU</i>	200	Diplomat
	400	Business
	700	Vacation
<i>Summary for 'R_ID' = EU (3 detail records)</i>		
Sum	1300	
<i>LA</i>	25	Diplomat
	50	Business
<i>Summary for 'R_ID' = LA (2 detail records)</i>		
Sum	75	
<i>ME</i>	10	Diplomat
	20	Business
<i>Summary for 'R_ID' = ME (2 detail records)</i>		
Sum	30	
<i>NA</i>		
<i>R_ID</i>	<i>Visit</i>	<i>Type</i>
	15	Business
	250	Diplomat
	700	Vacation
<i>Summary for 'R_ID' = NA (3 detail records)</i>		
<i>Grand Total 6430</i>		

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State the following:

- (i) The field used to group the records
 - (ii) The field used to sort the records
 - (iii) ONE other feature that was used in the report (3 marks)
- (g) You wish to add a field name **SFEE** to the VISIT table to calculate the security tax to be paid by the visitors. The security tax is computed by multiplying the number of visits by \$10. List the steps you would take to add the **SFEE** field to the VISIT table. (3 marks)
- (h) The result of a query is given below:

SumOfVisit
6430

- (i) Which table is used to perform the query?
 - (ii) What does 6430 represent? (2 marks)
- (i) The following query was created on the database:

Field:	R_ID	Type	Visit
Table:	visit	visit	visit
Sort:			
Show:	☒	☒	☒
Criteria:		"Vacation"	>500
or:			

- (i) What is the query doing? (2 marks)
- (ii) How many records will be displayed when the query is executed? (1 mark)

Total 20 marks

4. PROGRAMMING AND PROBLEM SOLVING

Consider the algorithm below.

Line Number

```
1
2      Number :=LIST[1]
3      For count = 2 to 6
4          Begin
5              If Number is greater than LIST
                [count]
6                  Then Number = LIST [count];
7          End For
8      output (Number);
9
```

- (a) Identify the variables in the algorithm. (2 marks)
- (b) Explain the purpose of the code in Line 2. (2 marks)
- (c) Write a statement in Pascal to declare the array. (3 marks)
- (d) Copy and complete the following trace table if the numbers 14, 15, 6, 9, 12, 4 are stored in the array.

Count	LIST[Count]	Number
1	14	14
2	15	
3	6	
4	9	
5	12	
6	4	

(4 marks)

- (e) For EACH of the following, write 'Yes' and the line number(s) if the term is found in the algorithm above otherwise write 'No':
 - (i) A logic operation
 - (ii) An arithmetic operation
 - (iii) A conditional statement
 - (iv) A loop (6 marks)
- (f) State the purpose of the algorithm. (2 marks)
- (g) Convert the corrected algorithm to a program written using Pascal. (11 marks)

Total 30 marks

END OF TEST